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Skin Electrical Impedance Model for Evaluation of the Thickness and Water Content of the Stratum Corneum

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Abstract Deterioration of the skin barrier function causes symptoms such as allergies because various chemical substances may enter the human body. Quantitative evaluation of the thickness and water content of the stratum corneum is useful as a measure of the skin barrier function in domains such as dermatology, nursing science, and cosmetics development. The stratum corneum is responsible for most of the skin barrier function, and two factors—the thickness and water content of the stratum corneum—are thus important. In this paper, the stratum corneum is regarded as a parallel model of resistance and capacitance. From measurements of the electrical impedance of the skin, we propose a new model for simultaneous estimation of the thickness and water content of the stratum corneum conventionally measured by a confocal laser scanning microscope and a confocal Raman spectrometer, respectively, and we discuss the results of the measurements. The electrical impedance of the skin was measured using a device that we developed. The measurement began 3 seconds after the electrodes on the measurement head of the device came into contact with the skin, and parameters including the impedance, which was obtained by applying an alternating current signal at two frequencies, were measured. We measured the thickness and water content of the stratum corneum using confocal laser microscopy and confocal Raman spectroscopy, respectively; investigated the relationship of the thickness and water content of the stratum corneum with the electrical impedance of the skin; and established a new potential model for estimating the thickness and water content of the stratum corneum from the parallel resistance and capacitance. The correlation coefficients of the verification data were 0.931 and 0.776, respectively; and the root-mean-squared error of the thickness of the stratum corneum was 2.3 μm, while the root-mean-squared error of the water content at the surface of the stratum corneum was 5.4 points. These findings indicate the feasibility of quantitative evaluation of the thickness and water content of the stratum corneum by measuring skin electrical impedance.

Keywords: skin electrical impedance, thickness, water content, stratum corneum, skin barrier.

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1. Introduction

A decline in the skin barrier function makes it easier for various chemicals to enter the body and cause allergies and other problems [1–3]. The skin barrier function is assessed in various fields including dermatology, nursing, and cosmetics development [4, 5]. Quantitative assessment of the skin barrier function is useful in many areas of research.

The skin barrier function has conventionally been evaluated using the transepidermal water loss (TEWL) [6]. The TEWL is measured on wiped skin after 15 min of acclimation at room temperature (21°C, 50% relative humidity) in a thermohydrostatic chamber. This method requires a temperature of 22°C or lower, which is unlikely to cause physiological perspiration, and should ideally be conducted in a facility environment where the

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temperature and humidity are controlled year round. However, it is difficult to control the effect of perspiration owing to the variable external environment, which causes variability in TEWL measurements.

Although the skin barrier function can be assessed by measuring the TEWL, it is unclear what in the stratum corneum affects the TEWL. The stratum corneum is considered to be responsible for most of the skin barrier function, and two factors, namely the thickness and water content of the stratum corneum, are thus important [7–9]. The thickness of the stratum corneum is related to the reciprocal of the TEWL, with a reported correlation coefficient of approximately 0.88 [9]. Therefore, the thickness of the stratum corneum is important in determining the skin barrier function. As an example, correlation of the severity of atopic dermatitis with the thickness and water content of the stratum corneum have been demonstrated [10]. Measurements of the thickness and water content of the stratum corneum may therefore be useful as objective biomarkers for the evaluation of the status of atopic dermatitis control.

However, there are no simple methods for quantifying the thickness and water content of the stratum corneum. In this paper, we propose a new theory adopting the electrical impedance method (hereinafter called the IM) for simultaneous estimation of the thickness and water content of the stratum corneum quantitively for evaluation of the skin barrier function.

The thickness of the stratum corneum can be measured using a confocal laser microscope [11]. We use a confocal laser microscope (VivaScope 1500, Caliber I.D. Inc., New York, USA) that takes cross-sectional images of the surface of the skin when the measuring probe is pressed against the skin. The cross-sectional images are then analyzed to calculate the thickness of the stratum corneum.

The water content of the stratum corneum can be measured using a confocal Raman spectrometer [12]. We use a confocal Raman spectrometer (gen2-SCA, River Diagnostics B.V., Rotterdam, Netherlands) that measures the moisture profile of the stratum corneum when the measuring stage is pressed against the skin. The water content of the stratum corneum is then calculated from the moisture profile.

However, both measurement of the stratum corneum thickness using a confocal laser microscope and measurement of the stratum corneum water content using a confocal Raman spectrometer are time consuming in terms of the measurements and data or image analysis, and they cannot be performed easily. Estimation of the thickness and water content of the stratum corneum using the IM has been studied as a solution to this problem.

The IM is a safe, non-invasive measurement method that has attracted much attention for the ease of measurement. Estimation of the stratum corneum moisture from the frequency characteristics of the electrical impedance of biological tissues has been widely adopted in, for example, the development of cosmetics. Values that correlate with the barrier function (TEWL values) can be calculated from impedance ratios derived from two alternating current (AC) signals in the measurement of the impedance of tape-stripped skin and normal skin. However, only a weak correlation has been observed, and it is not yet possible to precisely estimate the TEWL [13, 14]. In the case of IM, calculation of the thickness of the stratum corneum from measurements of skin impedance using a device called SciBase II has been reported [15]. However, many basic issues, including the lack of comparison with the measured thickness of the stratum corneum, remain. In this paper, we propose a new theory using IM for simultaneous estimation of the stratum corneum thickness and water content conventionally measured by confocal laser microscopy and confocal Raman spectroscopy, respectively, and discuss the results of the measurements. The data can subsequently be used for quantitative evaluation of the skin barrier function.

2. Theory

The theory of estimating the thickness (T) and water content (WC) of the stratum corneum from skin impedance is described here.

2.1 Modeling of the stratum corneum using IM

The impedance of the stratum corneum is represented by the RC parallel circuit shown in **Fig. 1** [16]. The parallel resistance R_P and its resistivity ρ, and the parallel capacitance C_P and its dielectric constant ε in this circuit are related by:

$$R_P = \rho \frac{T}{S} \, [\Omega] \,, \tag{1}$$

$$C_P = \varepsilon \frac{S}{T} \, [\mathrm{F}] \,, \tag{2}$$

where ρ is the resistivity, $S \left[\mathrm{m}^2\right]$ is the cross-sectional area of the stratum corneum, ε is the dielectric constant, and T [m] is the thickness of the stratum corneum.

The resistivity ρ and dielectric constant ε are affected by the water content of the stratum corneum. The fre-

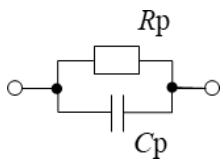


Fig. 1 RC parallel model of the impedance of the stratum corneum.

quency characteristics of the admittance (Y) of the skin, which is the reciprocal of the impedance (Z), for varying water content of the forearm epidermis have been compared with those of normal skin, revealing that skin with higher water content in the forearm epidermis has higher Y [17].

When tape stripping is performed to increase the water content of the stratum corneum. Due to the effect of the water content in the stratum corneum, there is an exponential increase in Y relative to that of normal skin. We thus suggest that the resistivity ρ and dielectric constant ε change exponentially with the water content of the stratum corneum. We propose a model in which the resistivity ρ decreases exponentially and the dielectric constant ε increases exponentially as described by

$$\rho = k_{\mathrm{R}} e^{-\beta_{\mathrm{R}} WC}\,[\Omega \cdot \mathrm{m}]\,, \tag{3}$$

$$\varepsilon = k_{\mathrm{C}} e^{\beta_{\mathrm{C}} WC}\,[\mathrm{F/m}]\,, \tag{4}$$

where k_{R} $[\Omega \cdot \mathrm{m}]$ and k_{C} $[\mathrm{F/m}]$ are parameters that are not related to the water content of the stratum corneum, and β_{R} $[1/\mathrm{mass} - \%]$ and β_{C} $[1/\mathrm{mass} - \%]$ are water content coefficients. Large β_{R} and β_{C} values respectively indicate small resistivity ρ and large dielectric constant ε.

Furthermore, as shown by Equations (1) and (2), R_{P} is proportional to the thickness of the stratum corneum while C_{P} is inversely proportional to the thickness of the stratum corneum. The resistivity ρ and dielectric constant ε are affected by the thickness of the stratum corneum [18]. The resistivity ρ decreases and the dielectric constant ε increases appreciably in deeper layers of the stratum corneum because of the laminated structure.

We therefore propose an exponential model that expresses the change in the distribution of resistivity in the thickness direction with R_{P} proportional to the thickness of the stratum corneum with a single exponential function. We also propose an exponential model that expresses the change in the dielectric constant distribution in the thickness direction with C_{P} inversely proportional to the thickness of the stratum corneum from a single exponential function. After applying these models, k_{R} and k_{C} can be expressed as:

$$k_{\mathrm{R}} = k_{\mathrm{R}}' T^{\gamma_{\mathrm{R}}-1}\,[\Omega \cdot \mathrm{m}]\,, \tag{5}$$

$$k_{\mathrm{C}} = k_{\mathrm{C}}' T^{-\gamma_{\mathrm{C}}+1}\,[\mathrm{F/m}]\,, \tag{6}$$

where $k_{\mathrm{R}}'\left[\Omega \cdot \mathrm{m}^{-\gamma_{\mathrm{R}}+2}\right]$ and $k_{\mathrm{C}}'\left[\mathrm{F/m}^{-\gamma_{\mathrm{C}}+2}\right]$ are constants that do not depend on the stratum corneum thickness or water content, and γ_{R} and γ_{C} are parameters indicating the non-uniformity of the resistivity in the direction of the thickness of the stratum corneum. If $\gamma_{\mathrm{R}} = \gamma_{\mathrm{C}} = 1$, k_{R} $[\Omega \cdot \mathrm{m}]$ and k_{C} $[\mathrm{F/m}]$ are uniform regardless of the thickness of the stratum corneum. Substitution of Equations (5) and (6) into Equations (3) and (4) gives:

$$\rho = k_{\mathrm{R}}' \cdot T^{\gamma_{\mathrm{R}}-1} \cdot e^{-\beta_{\mathrm{R}} WC}\,[\Omega \cdot \mathrm{m}]\,, \tag{7}$$

$$\varepsilon = k_{\mathrm{C}}' \cdot T^{-\gamma_{\mathrm{C}}+1} \cdot e^{\beta_{\mathrm{C}} WC}\,[\mathrm{F/m}]\,. \tag{8}$$

Substitution of Equations (7) and (8) into Equations (1) and (2), using the thickness and water content of the stratum corneum, gives:

$$R_{\mathrm{P}} = \alpha_{\mathrm{R}} \cdot T^{\gamma_{\mathrm{R}}} \cdot e^{-\beta_{\mathrm{R}} WC}\,[\Omega]\,, \tag{9}$$

$$C_{\mathrm{P}} = \frac{\alpha_{\mathrm{C}}}{T^{\gamma_{\mathrm{C}}}} e^{\beta_{\mathrm{C}} WC}\,[\mathrm{F}]\,, \tag{10}$$

where $\alpha_{\mathrm{R}} = \frac{k_{\mathrm{R}}'}{S}\,[\Omega \cdot \mathrm{m}^{-\gamma_{\mathrm{R}}}]$ and $\alpha_{\mathrm{C}} = S \cdot k_{\mathrm{C}}'\,[\mathrm{F} \cdot \mathrm{m}^{\gamma_{\mathrm{C}}}]$. Solving Equations (9) and (10) for T and WC yields:

$$T = \left(\frac{R_{\mathrm{P}}}{\alpha_{\mathrm{R}}}\right)^{\frac{\beta_{\mathrm{C}}}{\gamma_{\mathrm{R}}\beta_{\mathrm{C}} - \gamma_{\mathrm{C}}\beta_{\mathrm{R}}}} \left(\frac{C_{\mathrm{P}}}{\alpha_{\mathrm{C}}}\right)^{\frac{\beta_{\mathrm{R}}}{\gamma_{\mathrm{R}}\beta_{\mathrm{C}} - \gamma_{\mathrm{C}}\beta_{\mathrm{R}}}}\,[\mathrm{m}]\,, \tag{11}$$

$$WC = \frac{1}{\gamma_{\mathrm{R}}\beta_{\mathrm{C}} - \gamma_{\mathrm{C}}\beta_{\mathrm{R}}} \ln\left\{\left(\frac{C_{\mathrm{P}}}{\alpha_{\mathrm{C}}}\right)^{\gamma_{\mathrm{R}}} \left(\frac{R_{\mathrm{P}}}{\alpha_{\mathrm{R}}}\right)^{\gamma_{\mathrm{C}}}\right\}\,[\mathrm{mass} - \%] \tag{12}$$

2.2 Modeling of the stratum corneum with two frequencies using IM

A stratum corneum with a disrupted skin barrier function tends to have lower quantities of intercellular lipids that fill the gaps between keratinocytes, or tends to have a thinner stratum corneum [8, 9].

Impedance is obtained from a detection signal that is passed through the skin by applying an AC signal to normal skin and rough skin produced by tape stripping, and the detected admittance (Y) has different frequency characteristics for the two types of skin [17]. In the high-frequency region, the difference in Y between normal skin and rough skin produced by tape stripping is small and the values of Y can be considered the same, but the difference in Y is larger in the low-frequency region. The frequency characteristics of the skin impedance can be considered a factor for assessing the skin condition, and the impedance (Z), which is the reciprocal of Y, can thus also be considered a factor for evaluating the skin condition.

The AC signal in the low-frequency region flows through the intercellular lipids avoiding the keratinocytes because the cell membrane in the keratinocytes has high resistance. The cell membrane has high resistance but is thin and has a large electric capacity, and the AC signal in the high-frequency region can flow through regardless of the cell membrane. Using both the AC signal in the low-frequency region and the AC signal in the high-frequency region, it is possible to obtain information on both the keratinocytes and the intercellular lipids in the stratum corneum and to more accurately represent the skin condition.

We therefore calculate Z at high and low frequencies separately. Using both the high and low frequencies, we can more accurately calculate the thickness and water content of the stratum corneum.

R_{P}, C_{P}, α_{R}, α_{C}, β_{R}, β_{C}, γ_{R}, and γ_{C} respectively denote R_{PL}, C_{PL}, α_{RL}, α_{CL}, β_{RL}, β_{CL}, γ_{RL}, and γ_{CL} when using an AC signal in the low-frequency region; and R_{PH}, C_{PH},

α_{RH}, α_{CH}, β_{RH}, β_{CH}, γ_{RH}, and γ_{CH} when using an AC signal in the high-frequency region. Equation (9) can be rewritten as Equations (13) and (14) while Equation (10) can be rewritten as Equations (15) and (16). The high-frequency resistance R_{PH} is lower than the low-frequency resistance R_{PL}. The high-frequency capacitance C_{PH} is lower than the low-frequency capacitance C_{PL}. The equations are:

$$R_{PL} = \alpha_{RL} \cdot T^{\gamma_{RL}} \cdot e^{-\beta_{RL} WC} \, [\Omega] \,, \tag{13}$$

$$R_{PH} = \alpha_{RH} \cdot T^{\gamma_{RH}} \cdot e^{-\beta_{RH} WC} \, [\Omega] \,, \tag{14}$$

$$C_{PL} = \frac{\alpha_{CL}}{T^{\gamma_{CL}}} e^{\beta_{CL} WC} \, [\mathrm{F}] \,, \tag{15}$$

$$C_{PH} = \frac{\alpha_{CH}}{T^{\gamma_{CH}}} e^{\beta_{CH} WC} \, [\mathrm{F}] \,. \tag{16}$$

We obtain T and WC by selecting any two expressions from Equations (13) to (16) and creating a simultaneous expression. Among the six possible choices of combinations, we adopt the combination with the best coefficients of correlation between the estimated and measured values when calculating T and WC. The selected Equations (14) and (15) can be used to express T and WC as:

$$T = \left(\frac{R_{PH}}{\alpha_{RH}}\right)^{\frac{\beta_{CL}}{\gamma_{RH}\beta_{CL} - \gamma_{CL}\beta_{RH}}} \left(\frac{C_{PL}}{\alpha_{CL}}\right)^{\frac{\beta_{RH}}{\gamma_{RH}\beta_{CL} - \gamma_{CL}\beta_{RH}}} \, [\mathrm{m}] \,, \tag{17}$$

$$WC = \frac{1}{\gamma_{RH}\beta_{CL} - \gamma_{CL}\beta_{RH}} \ln \left\{ \left(\frac{C_{PL}}{\alpha_{CL}}\right)^{\gamma_{RH}} \left(\frac{R_{PH}}{\alpha_{RH}}\right)^{\gamma_{CL}} \right\} \, [\mathrm{mass} - \%] \,. \tag{18}$$

3. Measurement Methods

The following describes methods for measuring the electrical impedance of the skin, the water content of the stratum corneum, and the thickness of the stratum corneum.

Sixteen participants (15 males and one female, mean age of 42.1 ± 7.8 years) who did not have skin disease were included in this study. The test site was near the center of the medial side of the forearm (**Fig. 2**). Measurements were made from the normal skin in 16 participants and from the skin stripped with a circular tape 3 cm in diameter in 16 participants. The data were then divided into 16 cases for determining the constants in Equations (11) and (12) (hereafter referred to as fundamental data) and 16 cases for estimating T and WC using the determined constants (hereafter referred to as validation data).

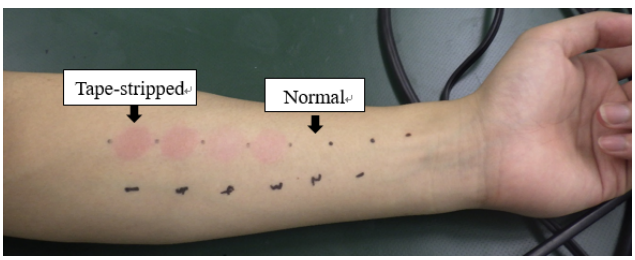


Fig. 2 Skin testing sites for measurement.

The impedance is obtained by detecting a signal that is passed through the skin after applying an AC signal, and the difference between the detected Y at 100 kHz and that at 3.5 MHz is thus not so large [17]. A frequency of 100 kHz is easy to use because there is unlikely to be noise at this frequency. We used a frequency of 100 kHz for the AC signal in the high-frequency region. When the frequency of the AC signal in the low-frequency region became low, the polarization impedance became large and difficult to measure. We adopted a frequency of 500 Hz for the AC signal in the low-frequency region so that the polarization impedance was relatively small and the measurement could be made easily.

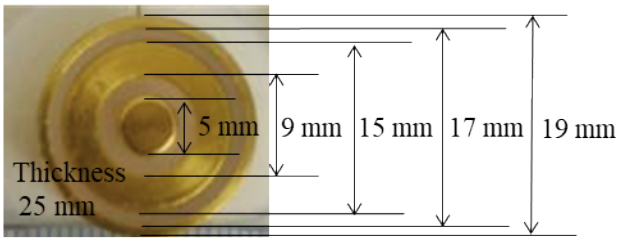
This study was conducted in accordance with the ethical principles of the Helsinki Declaration and after obtaining informed consent from each participant. The study was approved by the Ethical Review Committee of ALCARE Corporation (approval number DP04–48–20170929) and Okayama University Graduate School of Medicine, Dentistry and Pharmaceutical Sciences and Okayama University Hospital (approval number 2109–043). Measurements were made only after participants were informed about the experiment and their written consent was obtained.

3.1 Method of measuring the skin electrical impedance

The electrodes in the measuring head comprised a central circular sensing electrode (5 mm in diameter), a surrounding concentric ring-shaped applying electrode (9 mm in inner diameter, 15 mm in outer diameter), and an outermost ground electrode (17 mm in inner diameter, 19 mm in outer diameter) (**Fig. 3**). The ground electrode prevented external noise from affecting the detection signal. The electrodes in the measuring head were made of copper and gold plated. A current was passed through a saline solution from the electrodes of the skin electrical impedance measurement device and the depth of the current flow was confirmed. It was further confirmed that the current flow reached a depth of approximately 4 to 5 mm for both the 500-Hz and 100-kHz AC signals. Therefore, the measuring current would reach below the dermis layer with these electrodes. The measurement began 3 seconds after the head came into contact with the skin and was complete in approximately 5 seconds. The measurement of the skin electrical impedance is affected by the contact force of the skin, and the measuring de-

vice was thus equipped with a spring so that the force of contact with the skin was constant. A photograph of the skin electrical impedance measurement device in use is shown in **Fig. 4**.

Figure 5 is a block diagram of the skin electrical impedance measurement device that we fabricated. The AC voltage generated by the oscillating circuits of 500 Hz and 100 kHz is applied to the skin through the applying electrodes. The current passing through the skin is detected by the sensing electrode, and the detection unit amplifies the current signal passing through the skin and sends it to the arithmetic unit. The low-pass filter cuts high-frequency noise contained in the current signal passing through the skin. The cutoff frequency is 5.8 kHz for oscillation at 500 Hz and 230 kHz for oscillation at 100 kHz.



[Sensing electrode]
Central circular electrode
[Applying electrode]
Surrounding concentric ring-shaped electrode
[Ground electrode]
Outermost surrounding concentric ring-shaped electrode

Fig. 3 Electrodes in the measuring head.

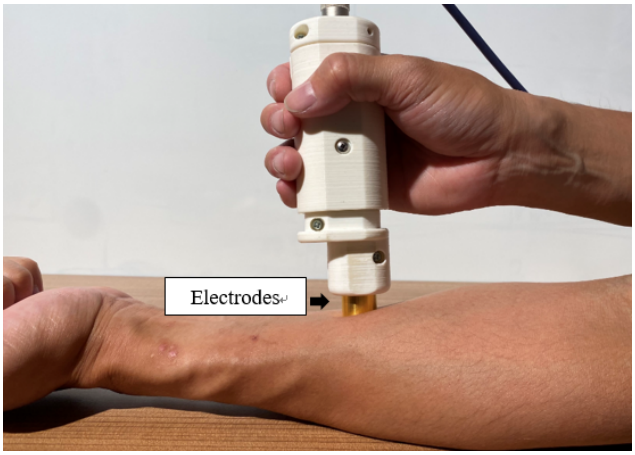


Fig. 4 Measurement of skin electrical impedance.

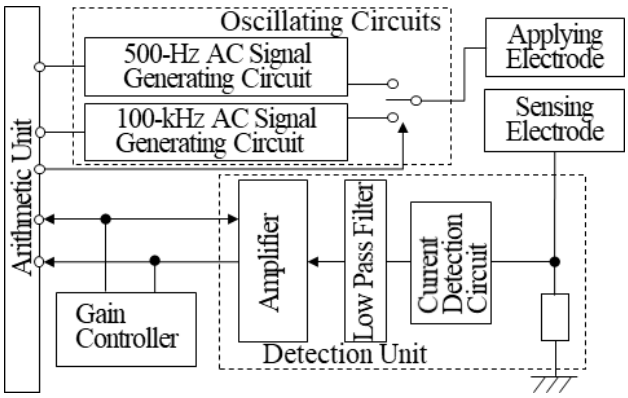


Fig. 5 Block diagram of the measurement device.

The gain controller detects the peak value of the detection signal and determines the gain of the amplifier. An analog–digital converter in the arithmetic unit digitally converts the amplified current signal. The sampling rate is 50 kS/s at 500 Hz and 10 MS/s at 100 kHz.

The arithmetic unit calculates the impedance using the peak voltage detected from the digitally converted current signal and the phase difference calculated from the time difference between the AC voltage and the current signal. The electrical impedance of the stratum corneum is then calculated by subtracting the impedance of tissue below the stratum corneum, such as that of the stratum granulosum.

The electrical impedance of the stratum corneum is expressed in Equation (19). R_P of the stratum corneum is calculated from the electrical impedance of the stratum corneum using Equation (20). C_P of the stratum corneum is calculated from the electrical impedance of the stratum corneum using Equation (21), where f denotes the frequency. The equations are:

$$Z = R + jX\,[\Omega]\,, \tag{19}$$

$$R_P = R + \frac{X^2}{R}\,[\Omega]\,, \tag{20}$$

$$C_P = \frac{X}{2\pi f\,(R^2 + X^2)}\,[\mathrm{F}]\,. \tag{21}$$

3.2 Method of measuring the water content of the stratum corneum

The water content of the stratum corneum was measured using a confocal Raman spectrometer (gen2-SCA). The measurement accuracy of the confocal Raman spectrometer is approximately ± 5 points to ± 8 points depending on the depth, and measurements can be made with good reproducibility [19]. In the present study, measurements were made at intervals of 2 μm from the surface of the stratum corneum to approximately 30 μm in the depth direction, after pressing the measurement stage against the test site. The water contents at different depths in the stratum corneum were averaged to obtain the water content of the stratum corneum. The signal collection time was 3 seconds per data point.

3.3 Method of measuring the thickness of the stratum corneum

The thickness of the stratum corneum was measured with a confocal laser microscope (VivaScope 1500). To measure the thickness of the stratum corneum using the confocal laser microscope, the probe was pressed against the skin and cross-sectional images of the surface of the skin were obtained and analyzed. In the analysis of cross-sectional images, we adopted the method of Integral Corporation (Tokyo, Japan) to determine the stratum corneum, using ripples generated at the interface between the skin and air at the crista cutis. The depth resolution of the cross-sectional images was 1.5 μm. The thickness of the stratum corneum was analyzed by taking two cross-sectional images in an area of 50 μm × 50 μm, and the average of the thicknesses of the two areas was calculated as T_M.

4. Analysis method

We define WC_{SM} as the average water content of the surface stratum corneum from the surface layer to 20% of T_M measured using a confocal Raman spectrometer. We define WC_{AM} as the average water content of the entire stratum corneum from the surface to T_M measured using a confocal Raman spectrometer. After calculating 20% of T_M measured by the confocal laser microscope, the average WC_{SM} was measured using the confocal Raman spectrometer.

The water content below the stratum granulosum is almost constant and high, and the water content of the stratum corneum decreases when approaching the surface layer [19]. It is difficult to capture differences in water content for the entire stratum corneum. We therefore focused on the water content from the surface layer to 20% of the thickness, where a difference in water content of the stratum corneum presumably exists.

Partial correlation coefficients in the multiple regression analysis are calculated to compare the effects of WC_{SM} and WC_{AM} on R_P and C_P.

Multiple regression analysis for WC_{SM} is performed by transforming both sides of Equations (7) to (10) to the natural logarithm form. In Equation (7), $\ln(R_{PL})$ is the objective variable while $\ln(T_M)$ and WC_{SM} are the explanatory variables. Equations (8) to (10) are processed in the same manner. In the multiple regression analysis of WC_{AM}, $\ln(T_M)$ and WC_{AM} are the explanatory variables.

In the determination of coefficients in equations (11) and (12), the KH method [20] is used for numerical calculations. The optimization condition is set to be the total of the sum of the squares of the residual rates of T, WC, R_{PH}, and C_{PL}. Residual rates are defined by:

$$\text{Residualrate} = (\text{estimatedvalue} - \text{measuredvalue}) / \text{measuredvalue}. \quad (22)$$

The constants obtained from the fundamental data are then used to estimate T and WC in the validation data.

5. Results

5.1 Comparison of the effects of water contents of the surface of stratum corneum and the entire stratum corneum on R_P and C_P

Both sides of Equations (7) to (10) were transformed to the natural logarithm form, and partial correlation coefficients were derived for the objective variables $\ln(R_{PL})$ and $\ln(C_{PH})$ and the explanatory variables WC and $\ln(T)$. The analysis was carried out for both WC_{SM} and WC_{AM}. **Table 1** shows the partial correlation coefficients for WC_{SM} and WC_{AM}.

There were no significant differences in the partial correlation coefficients between WC_{SM} and WC_{AM}. We thus confirmed that the different water contents of the stratum corneum; namely, WC_{SM} and WC_{AM}, have the same degree of effect on R_P and C_P.

5.2 Results of analysis of the fundamental data

Using the fundamental data, we estimated the thickness of the stratum corneum and water content of the surface stratum corneum. The impedance was measured with the skin electrical impedance measurement device. The estimated stratum corneum thickness is denoted T_E, while the estimated water content of the surface stratum corneum is denoted WC_{SE}.

The relationship between the estimated T_E and measured T_M is shown in **Fig. 6**, while the relationship between the estimated WC_{SE} and measured WC_{SM} is shown in **Fig. 7**; the correlation coefficients were 0.895 and 0.839, respectively, indicating high correlation.

Similarly, using the fundamental data, T_M and WC_{AM} were estimated. The relationship between the estimated T_E and measured T_M is shown in **Fig. 8** while the rela-

Table 1 Comparison of the partial correlation coefficients ($P < 0.005$).

	$\ln(R_{PL})$	$\ln(C_{PL})$	$\ln(R_{PH})$	$\ln(C_{PH})$
WC_{SM}	−0.546	0.529	−0.518	0.480
WC_{AM}	−0.573	0.555	−0.552	0.504

	$\ln(R_{PL})$	$\ln(C_{PL})$	$\ln(R_{PH})$	$\ln(C_{PH})$
$\ln(T)^*$	0.816	−0.822	0.771	−0.742
$\ln(T)^{**}$	0.914	−0.916	0.891	−0.871

*: for WC_{SM}, **: for WC_{AM}

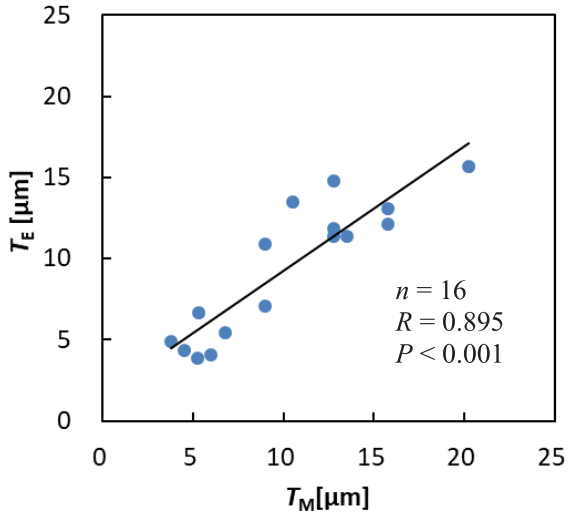


Fig. 6 Estimation of T_M using the fundamental data (for WC_{SM}). T_E: Ave. = 9.4 μm, S.D. = 4.1 μm, T_M: Ave. = 10.3 μm, S.D. = 4.8 μm.

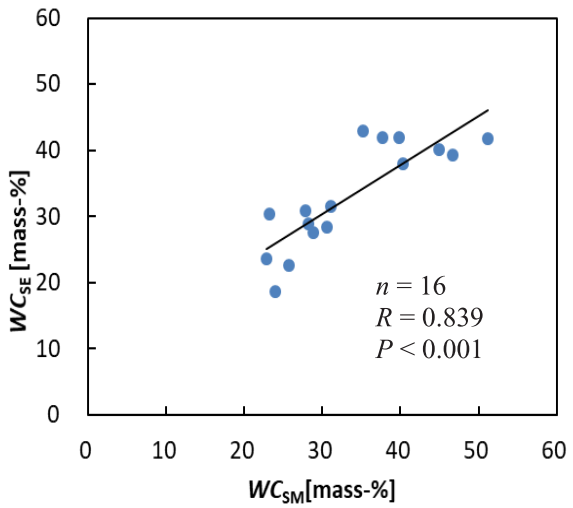


Fig. 7 Estimation of WC_{SM} using the fundamental data. WC_{SE}: Ave. = 33.0 mass-%, S.D. = 7.9 mass-%, WC_{SM}: Ave. = 33.6 mass-%, S.D. = 8.9 mass-%.

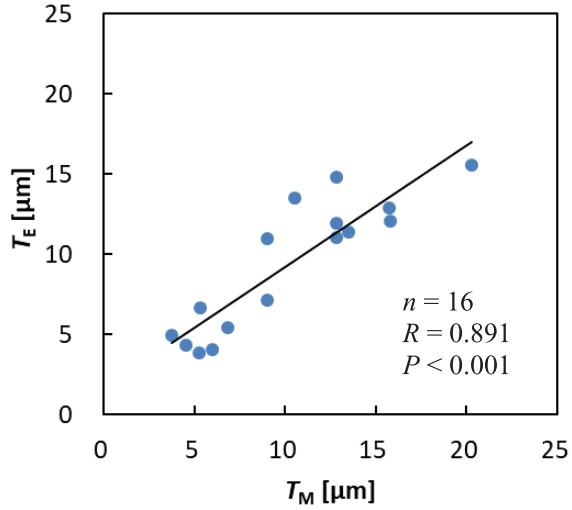


Fig. 8 Estimation of T_M using the fundamental data (for WC_{AM}). T_E: Ave. = 9.4 μm, S.D. = 4.1 μm, T_M: Ave. = 10.3 μm, S.D. = 4.8 μm.

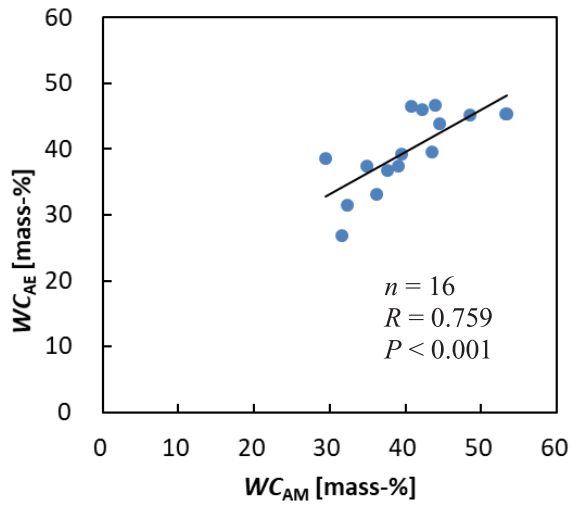


Fig. 9 Estimation of WC_{AM} using the fundamental data. WC_{AE}: Ave. = 39.9 mass-%, S.D. = 6.0 mass-%, WC_{AM}: Ave. = 40.7 mass-%, S.D. = 7.1 mass-%.

tionship between the estimated WC_{AE} and measured WC_{AM} is shown in **Fig. 9**; the correlation coefficients were 0.891 and 0.759, respectively, indicating high correlation. The correlation coefficients of T_E were not appreciably different when using WC_{SM} and WC_{AM}, but WC_{SE} was found to be more valid for use in an estimation model.

5.3 Results of analysis of the validation data

The determined coefficients were used to estimate T_M and WC_{SM} using the validation data. The relationship between the estimated T_E and measured T_M is shown in **Fig. 10** while the relationship between the estimated WC_{SE} and measured WC_{SM} is shown in **Fig. 11**; the correlation coefficients were 0.931 and 0.776, respectively, indicating high correlation.

6. Discussion

A new model for estimating the thickness and water content of the stratum corneum from R_P and C_P was found to be valid by comparing the stratum corneum thickness and water content estimated from the model with those measured by confocal laser microscopy and confocal Raman spectroscopy, respectively, and analyzing the relationship between the water content of the stratum corneum and the electrical impedance of the skin. There is

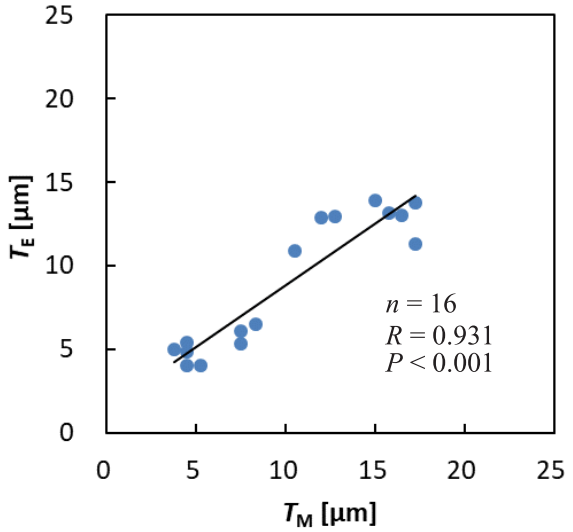


Fig. 10 Estimation of T_M using the validation data (for WC_{SM}). T_E: Ave. = 6.4 μm, S.D. = 5.0 μm, T_M: Ave. = 10.2 μm, S.D. = 5.1 μm.

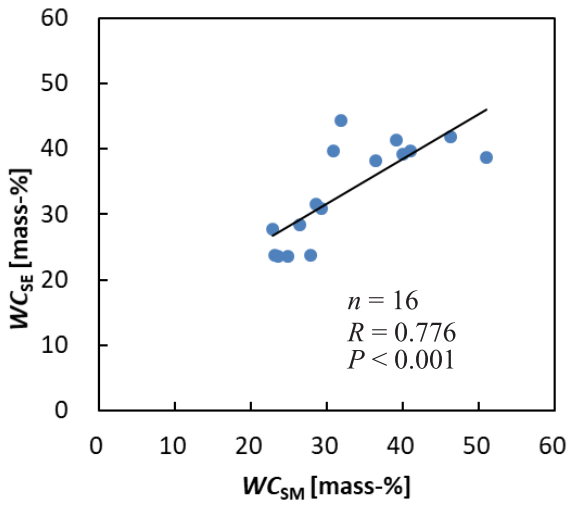


Fig. 11 Estimation of WC_{SM} using the validation data. WC_{SE}: Ave. = 22.5 mass-%, S.D. = 15.3 mass-%, WC_{SM}: Ave. = 32.7 mass-%, S.D. = 8.7 mass-%.

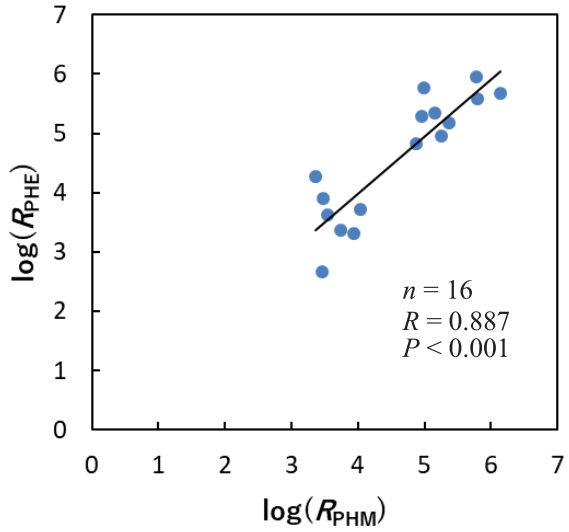


Fig. 12 Estimation of log(R_{PHM}) using the fundamental data.

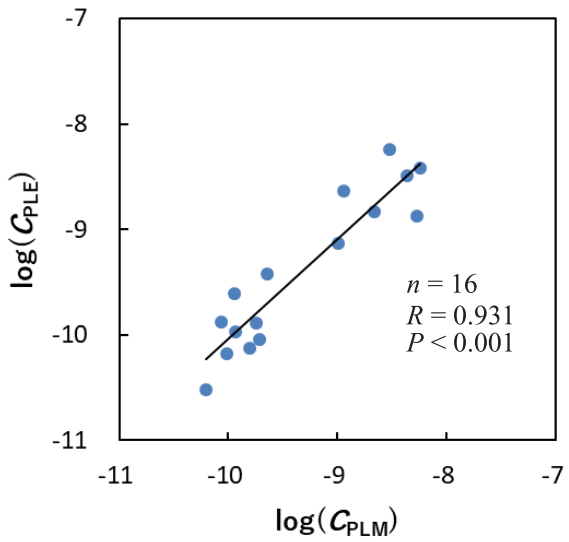


Fig. 13 Estimation of log(C_{PLM}) using the fundamental data.

no existing model for estimating the stratum corneum thickness conventionally measured by confocal laser microscopy and the stratum corneum water content conventionally measured by confocal Raman spectroscopy. The proposed model is thus expected to be a simple model that can be used to capture the state of the stratum corneum.

To verify the proposed model, both sides of Equations (9) and (10) were transformed to the common logarithm form. The values of log(R_{PH}) and log(C_{PL}) were estimated by substituting into Equations (9) and (10) the thickness of the stratum corneum T_E and the water content of the surface stratum corneum WC_{SE} estimated using the fundamental data.

Measurements obtained using the skin electrical impedance measurement device are denoted by R_{PHM} and C_{PLM}. Estimations obtained by substituting into equations (9) and (10) the thickness of the stratum corneum T_E and the water content of surface stratum corneum WC_{SE} using the fundamental data are denoted by R_{PHE} and C_{PLE}. The relationship between the estimated log(R_{PHE}) and measured log(R_{PHM}) is shown in **Fig. 12**, while the relationship between the estimated log(C_{PLE}) and measured log(C_{PLM}) is shown in **Fig. 13**; the correlation coefficients were 0.887 and 0.931, respectively, in-

dicating high correlation.

The models of Equations (3) and (4) were valid because R_{PH} and C_{PL} could be estimated with a high coefficient of determination using T_E and WC_{SE}.

In modeling the stratum corneum using two frequencies of the AC signal, the average water content of the surface stratum corneum (within 20% of the surface) was found to provide a more valid estimation than the average water content of the entire stratum corneum. This is because the dielectric constant ε and resistivity ρ are exponential functions of the moisture content, as shown in Equations (7) to (10), and the water content of the surface stratum corneum is therefore dominant in determining R_P and C_P.

T_E and WC_{SE} were estimated using the fundamental data. Results show a high correlation with T_M measured using a confocal laser microscope and WC_{SM} measured using a confocal Raman spectrometer. This provides a basis for the validity of the model described by Equations (3) and (4).

The combination of Equations (8) and (9) has highest correlation. This may be because the frequency-specific error components in R_{PH} and C_{PL} are not emphasized.

At present, the thickness and water content of the stratum corneum are not regularly measurable in clinical practice; therefore their values cannot be obtained with the degree of accuracy required for medical purpose. The target accuracy in estimating the thickness of the stratum corneum is 2 μm, such that the thickness of the stratum corneum of normal skin can be divided into approximately 10 steps with respect to the total stratum corneum thickness of 20 μm. The target accuracy in estimating the water content of the stratum corneum is 10 points, and the confocal Raman spectrometer thus has an error of ± 8 points as described above.

The measurement error ($|Z|$ is ± 5% and the phase difference is ± 3.6°) of the measurement circuit was confirmed. The effect on the estimated value was calculated to be 8.8% or less for the thickness of the stratum corneum and 3.2 points or less for the water content of the stratum corneum. When the equivalent circuit of the skin was determined using a device that measures the electrical impedance of the skin, the measurement error $|Z|$ was ± 2.5% and the phase difference was ± 3.3°. Additionally, it was confirmed that the measurement error was within the designed measurement error. The detection limit of the device is therefore 8.8% for the thickness of the stratum corneum and 3.2 points for the water content of the stratum corneum. The device that measures the electrical impedance of the skin has a limitation that it cannot capture changes smaller than these.

The root-mean-squared error (RMSE) of the thickness of the stratum corneum in the fundamental data was 2.2 μm. Considering that the positional accuracy of the confocal laser microscope was 1.5 μm in the z-direction, the residuals can be considered small. In the estimation of the water content of the surface stratum corneum, the RMSE was 4.7 points. Considering that the maximum variability of the confocal Raman spectrometer measurement was ± 8% [19], the residuals are considered small. These results of residuals confirm the appropriateness and accuracy of the model expressed as Equations (3) and (4).

The RMSE of the thickness of the stratum corneum in the verification data was 2.3 μm while the RMSE of the water content of the surface stratum corneum was 5.4 points. Even taking into account the measurement error of confocal laser microscopy and the measurement variability of confocal Raman spectroscopy, the residuals of the thickness of the stratum corneum and the water content of the surface stratum corneum in the data are considered small; the correlation coefficients of the validation data were high at 0.931 and 0.776, respectively. The coefficient of variation of the thickness of the stratum corneum was 0.47 when estimated from the skin electrical impedance measurement device and 0.50 when measured using confocal laser microscopy. The coefficient of variation of the water content of the surface stratum corneum was 0.23 when estimated from the skin electrical impedance measurement device and 0.27 when measured using confocal Raman spectrometry. Therefore, the device that measures the electrical impedance of the skin has precision similar to that of confocal laser microscopy and that of confocal Raman spectrometry.

According to the above results, the modeling proposed in this study has sensitivity, precision, and accuracy similar to those of a confocal laser scanning microscope and a confocal Raman spectrometer. It is thus possible to conduct qualitative evaluation at the present time.

The residuals when using the model proposed in this study may be due to the poor step width in the depth motion of the confocal laser microscope. Although a step width of 1.5 μm for the range of the stratum corneum is not sufficient, this is the limitation of current scientific capabilities. However, we expect more accurate estimation when the step width of the movement of the confocal laser microscope improves allowing measurement with a finer step width.

The measurement of the skin electrical impedance may be affected by the contact force of the skin, skin elasticity of the skin, measurement time, and sweat. To reduce the effect of the skin contact force, the measuring device is equipped with a spring so that the contact force of the skin is constant. When the skin is repeatedly mea-

sured, the elasticity of the skin decreases, which tends to lower the electrical impedance of the skin. Furthermore, when the measurement time exceeds approximately 30 seconds, the skin electrical impedance tends to decrease owing to sweating. Therefore, it is necessary to measure the skin electrical impedance so that the measurement can be completed in a short time without continuously measuring the electrical impedance.

When a current was passed through saline solution from the electrodes of the skin electrical impedance measurement device and the depth of the current flow was confirmed, both the 500-Hz and 100-kHz AC signals reached a depth of approximately 4 to 5 mm. There is no problem with the structure of the electrode because the AC signal reaches below the dermis layer when using this electrode.

The model of the stratum corneum in this study can be measured with the two frequencies of AC signals if the thickness of the stratum corneum is approximately 4 to 20 μm and the water content of the stratum corneum is 20 mass-% to 55 mass-%. This model may not be satisfactorily applicable for conditions outside these ranges.

The present study involved only 16 men and women as participants. As the number of people measured increases, there will be more data with stratum corneum thickness of 4 μm or less or 20 μm or more, and with stratum corneum water content of 20 mass-% or less or 55 mass-% or more. It is thus considered that the correlation coefficient will not be affected even if the number of people to be measured is increased. In future work, it will be necessary to increase the number of participants for measurement to confirm the results of the present study.

The model should be used with caution because the theoretical conditions of the model may not be met when the skin surface is coated with medications or when a disease such as atopic dermatitis develops.

In future work, we need to find signal frequencies that provide greater sensitivity to obtain higher correlation coefficients and to reduce the residuals. In addition, with the aim of applying the proposed method to skin diseases such as atopic dermatitis and senile dry skin, it is necessary to measure the skin under conditions other than those examined in this study.

7. Conclusion

We proposed a new modeling that calculates the thickness and water content of the stratum corneum simultaneously as measures of the functionality of the stratum corneum. Using a parallel circuit of resistance and capacitance that constitute the electrical impedance of the skin, we then established a potential model for estimating the thickness and water content of the stratum corneum that are conventionally measured by a confocal laser scanning microscope and a confocal Raman spectrometer, respectively. The results of all estimations were good, and it is thus expected that the model can be used as a simple model for capturing the state of the stratum corneum.

In addition, by measuring and quantifying the thickness and water content of the stratum corneum in approximately 5 seconds, it may be possible to obtain new evidence for diseases such as atopic dermatitis in the future.

Conflict of Interest

The authors declare no conflicts of interest with any companies or commercial organizations per the definition of the Japanese Society for Medical and Biological Engineering.

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